

The diagonal Amos-type family at real order: a machine-checked quantitative crossing classification

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Abstract

For the one-parameter family of Amos-type expressions

$$B_{\nu,c}(x) = \frac{x}{\nu + c + \sqrt{(\nu + c)^2 + x^2}}, \quad c \in \mathbb{R},$$

whose member $c = \frac{1}{2}$ is the classical Amos-type upper bound for the modified Bessel ratio $\rho_\nu = I_{\nu+1}/I_\nu$, we formalize in Lean 4, at every real order $\nu \geq 0$ over the Γ -power series, the classification of the parameter: $B_{\nu,c}$ is a uniform upper bound for ρ_ν (all $\nu \geq 0$, $x > 0$) exactly when $c \leq \frac{1}{2}$, and a uniform lower bound for every $c \geq 1$, with explicit counterexample witnesses. The classification statement itself is equivalent to known results of Ruiz-Antolín and Segura; for it we claim only the machine-checking. The contribution is the regime between the ends: for every $\nu \geq 0$ and every $c \in (\frac{1}{2}, 1)$ we prove that the fixed diagonal member $B_{\nu,c}$ crosses ρ_ν *exactly once* on $(0, \infty)$ — a transversal crossing located in the explicit window $[\sqrt{1-c}, 2\hat{D}/(2c-1)]$ with $\hat{D} = (\nu + \frac{3}{2})^2 - (\nu + c)^2$, obeying the strict threshold $x_* > x_\dagger = 2(\nu + c)\sqrt{c(1-c)}/(2c-1)$, with globally determined sign on both sides and the two-sided scale law $2(\nu + c)\sqrt{c(1-c)} \leq (2c-1)x_* \leq 2\hat{D}$. Degenerate (critical) contact is *excluded by theorem*, through the exact identity $(\nu + c)^2 D''(x_\dagger) = (2c-1)^3 B_{\nu,c}(x_\dagger)$ for the gap $D = \rho_\nu - B_{\nu,c}$ at any hypothetical critical contact. The entire chain carries the axiom oracle `[propext, Classical.choice, Quot.sound]` and no analytic hypothesis beyond $\nu \geq 0$, $x > 0$; a pre-registered certified interval-arithmetic companion verifies the crossing phenomenon independently of the C5 crossing theorems at 30 parameter pairs spanning the hard regimes, all passing at 128 bits.

1 Introduction

The ratio $\rho_\nu(x) = I_{\nu+1}(x)/I_\nu(x)$ of modified Bessel functions of the first kind admits algebraically calibrated bounds of the shape $x/(\lambda + \sqrt{\mu^2 + x^2})$; the literature on such bounds — Amos [1], Segura [2, 5, 6], Hornik–Grün [3, 7], Ruiz-Antolín–Segura [4] — is rich, and the one-parameter *diagonal* family studied here,

$$B_{\nu,c}(x) = \frac{x}{\nu + c + \sqrt{(\nu + c)^2 + x^2}},$$

*This manuscript and parts of the formal development were produced with AI assistance under the repository’s audit protocol. Every theorem below is machine-checked in Lean 4 against a pinned Mathlib; the independent numerical companion of Section 6 is a committed interval-arithmetic transcript; remaining numerics are labelled paper-level where they occur. Repository: <https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME>.

is, up to reparametrization, the family whose sharp ends [4] identifies: under $\mu = \nu + 1$, $\alpha = 2c - 1$, the family b_α of [4] is exactly $B_{\nu,c}$. *No novelty is claimed for the family, nor for the qualitative observation that its non-bounding members must cross the ratio* — [4] already notes that for parameters between the sharp ends the graphs cross. Three companion papers of this series formalized the consequence calculus of the $c = \frac{1}{2}$ bound at integer order [11], the bound itself at integer order [12], and the bound at every real order $\nu \geq 0$ over the Γ -power series [13]; the present paper builds directly on the last.

The contribution here is twofold, and the two halves carry different claims:

1. **The classification, machine-checked.** For every real $\nu \geq 0$ and $x > 0$: $\rho_\nu < B_{\nu,c}$ holds uniformly iff $c \leq \frac{1}{2}$ (Theorem 3.1), and for $c \geq 1$ the reverse inequality holds uniformly (Theorem 3.2). As mathematics this is equivalent to results of [4]; the claim is the formalization — uniform in real order, from the series, with explicit rational witnesses, inside one pinned axiom-clean development.
2. **The quantitative crossing theorem.** For $\nu \geq 0$ and $\frac{1}{2} < c < 1$, the fixed diagonal member $B_{\nu,c}$ crosses ρ_ν exactly once; the crossing is transversal, satisfies an explicit strict threshold, determines the sign of $\rho_\nu - B_{\nu,c}$ globally on both sides, lies in a constructive finite window, and obeys an explicit two-sided scale law (Theorem 3.3). Degenerate contact is excluded by an exact second-derivative identity (Theorem 3.4). We are not aware of a previous theorem giving this complete quantitative crossing classification for the diagonal family; Section 7 compares against the closest antecedents parameter by parameter.

Statuses: **exact** means a Lean theorem in the pinned central repository with clean axiom oracle; **certified** means a committed interval-arithmetic transcript; **paper-level** means ordinary prose, always flagged; **verified** marks an ordinary high-precision computation used only as a cross-check of a certified result, never as evidence. The repository’s largest namespace (`YangMills/`) reflects its original motivation in lattice gauge theory; no reduction to any continuum or gauge-theoretic problem is claimed or used.

2 Setting

All objects live in module `AmosClosure` of the pinned central repository, over the real-order interface of [13]: $I_\nu = \text{besselIReal } \nu$ is the Γ -power series $\sum_{k \geq 0} (x/2)^{\nu+2k} / (k! \Gamma(\nu + k + 1))$ (real exponents via `rpow`), with positivity, the three-term recurrence, termwise differentiability, the Riccati equation

$$\rho'_\nu(x) = Q_x(\rho_\nu(x)), \quad Q_x(y) := 1 - \frac{2\nu + 1}{x} y - y^2, \quad (1)$$

the barrier $\psi_\nu > 2\nu + 1$ for $\psi_\nu(x) = x(1/\rho_\nu - \rho_\nu)$, and the $c = \frac{1}{2}$ bound $\rho_\nu < B_{\nu,1/2}$ (`amosBoundReal_holds`). New objects of this paper: the family $B_{\nu,c} = \text{amosFamily } \nu \ c$, the lower bound $L_\nu = \text{amosLower } \nu$ of Section 4.2, and the crossing gap

$$D(x) = \rho_\nu(x) - B_{\nu,c}(x).$$

Throughout, $a = \nu + c$ and $s = \sqrt{a^2 + x^2}$. Everything is stated at $x > 0$, the `rpow` domain of the development.

3 Main results

Theorem 3.1 (classification of the upper-bound range; `amosFamily_uniform_upper_iff`). *For every $c \in \mathbb{R}$:*

$$\left(\forall \nu \geq 0, \forall x > 0: \rho_\nu(x) < B_{\nu,c}(x) \right) \iff c \leq \frac{1}{2}.$$

The failure witness for $c > \frac{1}{2}$ is explicit and rational: $x_0(c) = 2/(2c - 1)$ at $\nu = 0$ (`amosFamily_lt_amosLower_witness`).

Theorem 3.2 (the uniform lower half; `amosFamily_lower_of_one_le`). *For every $c \geq 1$, every $\nu \geq 0$ and every $x > 0$, $B_{\nu,c}(x) < \rho_\nu(x)$.*

Theorems 3.1 and 3.2 are machine-checked forms of known mathematics [4]; between them lies the regime $\frac{1}{2} < c < 1$, where $B_{\nu,c}$ bounds on neither side. There — with the strictness of the threshold in item (ii), and through it the unconditional uniqueness and orientation, delivered by Theorem 3.4 below:

Theorem 3.3 (the global crossing theorem; `amosFamily_global_crossing`, `amosFamily_crossing_orientation`). *Fix $\nu \geq 0$ and $\frac{1}{2} < c < 1$. There is exactly one $x_* > 0$ with $\rho_\nu(x_*) = B_{\nu,c}(x_*)$, and:*

- (i) (window) $\sqrt{1-c} \leq x_* \leq \frac{2\hat{D}}{2c-1}$, where $\hat{D} = (\nu + \frac{3}{2})^2 - (\nu + c)^2$ (`amosFamily_crossing_exists`);
- (ii) (strict threshold) $x_* > x_\dagger := \frac{2(\nu+c)\sqrt{c(1-c)}}{2c-1}$ (`crossing_threshold_strict`);
- (iii) (transversality) at the crossing, $D'(x_*) = \frac{B_{\nu,c}(x_*)}{x_*} \left(2c - 1 - \frac{\nu+c}{\sqrt{(\nu+c)^2 + x_*^2}} \right) > 0$ (`crossing_hasDerivAt`, whose factor is positive by `crossing_threshold_strict`);
- (iv) (orientation) $\rho_\nu < B_{\nu,c}$ on $(0, x_*)$ and $\rho_\nu > B_{\nu,c}$ on (x_*, ∞) (`amosFamily_crossing_orientation`); conversely — an immediate trichotomy consequence — the sign of D locates x relative to x_* ;
- (v) (scale law) $2(\nu+c)\sqrt{c(1-c)} \leq (2c-1)x_* \leq 2\hat{D}$ (`crossing_scale`).

Theorem 3.4 (no degenerate contact; `crossing_no_critical_contact`). *Fix $\nu \geq 0$ and $\frac{1}{2} < c < 1$. No crossing of ρ_ν and $B_{\nu,c}$ is a critical point of D : at any hypothetical contact with $D = D' = 0$ — which the slope identity forces to lie exactly at $x_\dagger$ — the second derivative obeys the exact identity*

$$(\nu+c)^2 D''(x_\dagger) = (2c-1)^3 B_{\nu,c}(x_\dagger) > 0,$$

and a critical contact with positive curvature contradicts the below-threshold orientation. Consequently the threshold in Theorem 3.3 is strict and uniqueness and orientation hold unconditionally.

Theorem 3.5 (the Turán inequality at real order; `besselIReal_turan`). *For every $\nu \geq 0$ and $x > 0$, $I_{\nu+1}(x)^2 > I_\nu(x) I_{\nu+2}(x)$; equivalently $\psi_\nu(x) < 2\nu + 2$, completing the sandwich $2\nu + 1 < \psi_\nu(x) < 2\nu + 2$ whose lower half is the barrier of [13]. This inequality is classical (Turán-type inequalities for I_ν go back to Thiruvengatachar–Nanjundiah; see Baricz–Ponnusamy [9]); it is proved here at real order over the Γ -series and used as supporting structure.*

4 The proof

The formal proofs are the authority; this section presents the mathematics in full, at the level of the checklist a referee needs.

4.1 Family algebra: calibration and the nullcline trichotomy

Since $s = \sqrt{a^2 + x^2} > |a| \geq -a$, the denominator $a + s$ is positive for *every* real c , and the conjugate form $B_{\nu,c} = (s - a)/x$ follows from $(s - a)(s + a) = x^2$ (`amosFamily_pos`). The general calibration identity

$$\frac{1}{B_{\nu,c}(x)} - B_{\nu,c}(x) = \frac{2(\nu + c)}{x} \quad (\text{amosFamily_calibration}) \quad (2)$$

extends the $c = \frac{1}{2}$ calibration of [13]. Substituting into the Riccati quadratic of (1): $1 - B^2 = (2a/x)B$, so

$$Q_x(B_{\nu,c}(x)) = \frac{2a - (2\nu + 1)}{x} B_{\nu,c}(x) = \frac{2c - 1}{x} B_{\nu,c}(x) \quad (\text{riccatiQReal_amosFamily}), \quad (3)$$

with the sign trichotomy: zero iff $c = \frac{1}{2}$ (the member *on* the nullcline — exactly the calibration that drives the proof of the $c = \frac{1}{2}$ bound), positive iff $c > \frac{1}{2}$, negative iff $c < \frac{1}{2}$. Since $a \mapsto x/(a + \sqrt{a^2 + x^2})$ is decreasing (`amosFamily_anti`; the non-strict form is all the chain uses), $c \leq \frac{1}{2}$ gives $B_{\nu,c} \geq B_{\nu,1/2} > \rho_\nu$ by [13]: the sufficiency half of Theorem 3.1 (`amosFamily_upper_of_le_half`).

4.2 The lower bound by recurrence inversion

The ratio recurrence $1/\rho_\nu = 2(\nu + 1)/x + \rho_{\nu+1}$ (`besselIReal_ratio_recurrence`) turns the $c = \frac{1}{2}$ *upper* bound at order $\nu + 1$ into a *lower* bound at order ν : with $b = \nu + \frac{3}{2}$ and $s' = \sqrt{b^2 + x^2}$,

$$\frac{1}{\rho_\nu(x)} < \frac{2(\nu + 1)}{x} + \frac{s' - b}{x} = \frac{\nu + \frac{1}{2} + s'}{x}, \quad \text{i.e.} \quad \rho_\nu(x) > \frac{x}{\nu + \frac{1}{2} + \sqrt{(\nu + \frac{3}{2})^2 + x^2}} =: L_\nu(x),$$

where the middle step is the algebraic collapse $x^2/(b + s') = s' - b$ (`besselLowerReal_holds`). This is the sharp-at-infinity lower bound of [4]; here it is a three-line corollary of the real-order upper bound, which is the point of recording it. L_ν matches ρ_ν to first order at infinity ($L_\nu \sim 1 - (\nu + \frac{1}{2})/x$), while $B_{\nu,c} \sim 1 - (\nu + c)/x$ (paper-level asymptotics, not used in the proofs); for $c > \frac{1}{2}$ the family member falls below L_ν at explicit finite x : at $\nu = 0$ and $x_0(c) = 2/(2c - 1)$ the inequality $B_{0,c}(x_0) < L_0(x_0)$ reduces, after clearing the calibrated denominators with $(2c - 1)x_0 = 2$, to the rational inequality $\frac{9}{4} - c^2 < 2$ for $c \in (\frac{1}{2}, 1]$ and its routine extension beyond (`amosFamily_lt_amosLower_witness`); chaining $B_{0,c}(x_0) < L_0(x_0) < \rho_0(x_0)$ kills uniformity. That is the necessity half of Theorem 3.1. For Theorem 3.2: $c \geq 1$ gives $B_{\nu,c} \leq B_{\nu,1} < L_\nu < \rho_\nu$, where $B_{\nu,1} < L_\nu$ is the root comparison $\sqrt{(\nu + \frac{3}{2})^2 + x^2} < \frac{1}{2} + \sqrt{(\nu + 1)^2 + x^2}$, i.e. $\nu + 1 < \sqrt{(\nu + 1)^2 + x^2}$ after squaring (`amosFamily_one_lt_amosLower`).

4.3 The Turán sandwich

$\rho_\nu > B_{\nu,1}$ (from Section 4.2) plus the calibration (2) at $c = 1$ gives $\psi_\nu < 2(\nu + 1)$ (`besselPsiReal_lt`); together with the barrier $\psi_\nu > 2\nu + 1$ of [13] this is the sandwich $2\nu + 1 < \psi_\nu < 2\nu + 2$. The upper half converts, through the recurrence, into $\rho_\nu > \rho_{\nu+1}$ and thence into the Turán inequality of Theorem 3.5 (`besselIReal_turan`). We emphasize provenance: the mathematical content of the sandwich upper half is classical [9]; the formal statement at real order over the Γ -series is what is new here, and the crossing results below do not depend on it.

4.4 Existence: the explicit window

Fix $\frac{1}{2} < c < 1$. At the left end, the product-form upper bound $\rho_\nu(x) < x/(2(\nu + 1)(1 - q_\nu))$ with $q_\nu = (x/2)^2/(\nu + 2)$, valid for $q_\nu < 1$ (`besselRatioReal_lt_product`, from the geometric-domination toolkit of [13]) beats the family member below $\sqrt{1 - c}$: using $\sqrt{a^2 + x^2} \leq a + x^2/(2a)$ one checks $D(x) < 0$ for all $0 < x \leq \sqrt{1 - c}$, uniformly in ν (`crossing_witness_small_le`). At the right end, at $X = 2\hat{D}/(2c - 1)$ with $\hat{D} = (\nu + \frac{3}{2})^2 - (\nu + c)^2$, the two calibrated roots separate by less than $(2c - 1)/4$ and $L_\nu(X) > B_{\nu,c}(X)$, so $D(X) > 0$ (`crossing_witness_large`). Continuity of D (from differentiability) and the intermediate value theorem give a crossing in the window (`amosFamily_crossing_exists`).

4.5 Slope identity, threshold, uniqueness, orientation

The family derivative is $B'_{\nu,c}(x) = a/(s(a + s))$ (`amosFamily_hasDerivAt`). At any zero of D , ρ_ν takes the family value, so by (1) and (3) $\rho'_\nu = \frac{2c-1}{x}B_{\nu,c}$ there, while $a/(s(a + s)) = aB_{\nu,c}/(sx)$ by the conjugate form; hence the *slope identity*

$$D'(\text{zero}) = \frac{B_{\nu,c}}{x} \left(2c - 1 - \frac{a}{s} \right) \quad (\text{crossing_hasDerivAt}). \quad (4)$$

The factor $2c - 1 - a/s$ is negative below, zero at, and positive above $x_\dagger = 2a\sqrt{c(1 - c)}/(2c - 1)$ (where $s = a/(2c - 1)$). Three consequences, in order: every crossing satisfies $x \geq x_\dagger$, since a crossing strictly below the threshold would be a strictly *downward* crossing yet $D < 0$ must persist from the left window endpoint (`crossing_threshold`); above the threshold every crossing is strictly upward (`crossing_pos_right`), and two strictly upward zeros of a differentiable function cannot occur without an intermediate downward zero — excluded — so the crossing is unique (`crossing_unique`); and the sign of D is then globally determined on both sides (`crossing_orientation_after`, `crossing_orientation_below`).

4.6 Exclusion of the degenerate contact

One case survives Section 4.5: a contact exactly at $x_\dagger$, where the slope factor vanishes identically and the crossing would be a critical point of D . At such a contact every quantity is an explicit algebraic function of (ν, c) : writing $t = 2c - 1 \in (0, 1)$,

$$s = \frac{a}{t}, \quad \rho_\nu = B_{\nu,c} = \sqrt{\frac{1 - c}{c}}, \quad B_{\nu,c}x = s - a = \frac{a(1 - t)}{t}, \quad x^2 = \frac{a^2(1 - t)(1 + t)}{t^2},$$

and $2\nu + 1 = 2a - t$. Differentiating the Riccati equation along the flow and the family derivative once more (`crossing_hasDerivAt_everywhere` provides D 's derivative as an explicit function

everywhere; that function is differentiated once more inside the tangency proof), the two curvatures collapse to

$$\rho''_\nu = \frac{B}{x^2}[(2\nu+1)(1-t)-2tBx] = -\frac{t^3 B}{a^2(1+t)}, \quad B'' = -\frac{a(x/s)(a+2s)}{s^2(a+s)^2} = -\frac{t^3 B(t+2)}{a^2(t+1)},$$

whence the exact identity of Theorem 3.4: $D'' = t^3 B/a^2$, i.e. $(\nu+c)^2 D'' = (2c-1)^3 B_{\nu,c} > 0$. In the formal development the identity is proved division-free in exactly this form; positivity needs only $2c-1 > 0$, $\nu+c > 0$, $B_{\nu,c} > 0$. The contradiction: at a critical contact, $D' = 0$ and $D'' > 0$ force $D' < 0$ on a left interval, so D is strictly decreasing into the contact and $D(y) > D(x_+) = 0$ just left of it; but $y < x_+$ is strictly below the threshold, where the orientation of Section 4.5 gives $D(y) < 0$ (`crossing_no_critical_contact`). No critical contact exists; the threshold is strict (`crossing_threshold_strict`); uniqueness and orientation are unconditional (`crossing_unique_unconditional`); and the packaged existence-and-uniqueness statement is `amosFamily_global_crossing`. The scale law of Theorem 3.3(v) is then immediate from the threshold and the window (`crossing_scale`).

5 Scope

All results are for the in-core Γ -power-series definition of I_ν at real order $\nu \geq 0$, $x > 0$; no identification with an external special-functions library object is claimed (none exists at the pinned Mathlib to identify with), though the identification theorem of [13] ties the object to the factorial series at every integer order. Theorems 3.1 and 3.2 are machine-checked forms of known mathematics [4], and the crossing *phenomenon* for intermediate parameters is likewise observed there; the claims of this paper are (i) the formalization, uniform in real order, and (ii) the quantitative crossing package of Theorems 3.3 and 3.4: unique transversal crossing, strict explicit threshold, bilateral orientation, constructive window, two-sided scale law, degenerate contact excluded. We are not aware of a previous theorem giving this complete quantitative crossing classification for the diagonal family.

6 The certified companion

A certified interval-arithmetic companion verifies the crossing phenomenon *independently of the C5 crossing theorems* — without consulting any C5 crossing, uniqueness, orientation, or scale-law theorem: its protocol — registered in the development’s charter in its own commits, strictly before the certified run and before any certified result was observed — mandates that verdicts be decided *only* by ball signs of D and D' , never by consulting the crossing theorems; that the bisection search interval come from the fixed rule $[(1-c)/4, 8(\nu+2)^2/(2c-1)]$, deliberately *wider* than the Lean window, with the theoretical window checked afterwards rather than searched; and that failures escalate a precision ladder (128 to 4096 bits) rather than widen intervals. The grid of 30 pairs, $\nu \in \{0, \frac{1}{2}, 1, \pi, 10, 100\} \times c \in \{0.501, 0.6, 0.75, 0.9, 0.999\}$ (π a certified ball), forces the hard regimes: $c \downarrow \frac{1}{2}$ (the crossing escapes to infinity — at $(\nu, c) = (100, 0.501)$ the search interval reaches 4.2×10^7), $c \uparrow 1$ (the crossing value $B(x_+) \rightarrow 0$), $\nu = 0$, large order, and the doubly-extreme corners. Per pair, four certificates, all by ball sign: a bracketing box $X = [\text{lo}, \text{hi}]$ with certified opposite signs of D at its ends (certified bisection; midpoints are outward-rounded balls — for non-dyadic parameters the registered exact-dyadic midpoints are not representable — a strictly conservative refinement, since wider endpoint balls make every

certificate harder, never easier); transversality $\inf_{x \in X} D'(x) > 0$ by interval evaluation over the hull; window inclusion $X \subset [x_+, x_+]$; and both sides of the scale law. The ratio is evaluated by the backward recurrence $\rho_\alpha = x/(2(\alpha+1) + x\rho_{\alpha+1})$ seeded at order $\nu + N$ with the *interval* $[0, 1]$ — valid by positivity and the $c = \frac{1}{2}$ bound of [13], both outside the forbidden theorem set — so no exponentially large value of I_ν is ever formed; the seed contraction is certified a posteriori by the final ball width, and a too-wide ball can only produce an undecided verdict, never a wrong one. All 30 pairs pass all four certificates at the first ladder rung (128 bits); an independent ordinary high-precision pass (labelled *verified*, not certified) agrees at all 30. Two illustrative rows: at $(100, 0.501)$ the certified box $[50250.8995 \dots]$ (width $\sim 3.5 \times 10^{-14}$) clears the threshold $x_+ = 50250.3995 \dots$ with certified D' ball $= 3.95 \times 10^{-13} \pm 4.1 \times 10^{-16} > 0$; at $(\frac{1}{2}, 0.501)$ the box sits at $x_* = 501.00000000000000 \dots$, reflecting the closed (hyperbolic-cotangent) form of $\rho_{1/2}$: by the calibration (2), the crossing equation is $\psi_{1/2}(x_*) = 2(\nu + c)$, and $\psi_{1/2}(x) = 2 + 1/(x - 1)$ up to exponentially small error, so $x_* \approx 1 + \frac{1}{2c-1} = \frac{2c}{2c-1} = 501$ — a paper-level remark; the certified box needs no such structure. The companion is *not* part of the proof, and the proof does not cite it.

7 Related work

The comparison below follows the development’s pre-registered collation; the antecedents are compared claim by claim.

Ruiz-Antolín–Segura [4] is the exact antecedent of the *family* and of *qualitative existence*: under $\mu = \nu + 1$, $\alpha = 2c - 1$, their b_α family is $B_{\nu,c}$; they identify the sharp ends ($\alpha \leq 0$, i.e. $c \leq \frac{1}{2}$, at $x \rightarrow \infty$; $\alpha \geq 1$, i.e. $c \geq 1$, at $x \rightarrow 0$) and observe that for intermediate parameters the graphs must cross (asymptotic end signs). We declare expressly: the family and the mere existence of some crossing are prior art. [4] contains no uniqueness, orientation, threshold, tangency exclusion, or quantitative localization for the intermediate members.

Hornik–Grün [3] is the closest antecedent in spirit: it characterizes which two-parameter Amos-type expressions $G_{\alpha,\beta}$ are upper or lower bounds, and describes minimal upper bounds as those tangent to the ratio at a single point. The statements are about the *boundary of the bounding region* in the (α, β) -plane; classifying the fixed diagonal $G_{\alpha,\alpha}$ -type member $B_{\nu,c}$ — which for $c \in (\frac{1}{2}, 1)$ is *not* a bound at all — is a different question. Parameter by parameter: [3] yields, for the diagonal, membership/non-membership in the bounding classes; it does not state, for non-bounding diagonal members, the unique simple crossing, its transversality, the explicit threshold x_+ , bilateral orientation, or a scale law.

Segura [6] studies bounds $(\alpha + \sqrt{\beta^2 + \gamma^2 x^2})/x$ -type with coefficients *optimized as functions of a prescribed tangency point*: there the bounding function is a global bound whose graph is tangent to the ratio (osculatory contact by construction). Here the family member is *fixed in advance*, is not a bound, the meeting is transversal, and tangential contact is *excluded by theorem*. Close in machinery, not equivalent in statement.

Hornik–Grün [7] proves single-sign-change results *between two bounds* within their generalized family (bound-versus-bound comparisons); the present single-crossing statement is *ratio-versus-family-member*, a different pair of curves — we flag the terminological near-collision explicitly.

Segura [5] (Riccati/nullcline monotonicity methods) and **Salazar** [8] (Bernstein-positivity certificates for Riccati reductions) supply methodology adjacent to ours; neither presents the quantitative diagonal crossing package.

On the formalization side, the companion papers [11, 12, 13] formalized the $c = \frac{1}{2}$ bound and its consumers; we know of no other machine-checked development of Amos-type bounds or of any crossing classification. The recurrence and derivative identities are [10, 10.29]. The development builds on Mathlib [14].

8 Reproducibility

Toolchain: Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit

07642720480157414db592fa85b626dafb71355b

(lakefile and manifest agree); `tectonic` 0.15.0; `python-flint` 0.9.0 for the certified companion.

Recorded build facts: at the final mathematics commit

`f1c87fef` (*short id; the release manifest records the full hashes*)

`lake build AmosClosure` completed successfully (**8175 jobs**); the axiom oracle printed, for all **110** registered statements (the 74 of the `c4-v1.0` release registry plus the 36 of this paper), exactly

`[propext, Classical.choice, Quot.sound].`

One proof-engineering disclosure: the division-free identity inside the tangency-exclusion proof (`crossing_no_critical_contact`) is closed by `linear_combination` with cofactor $K = -2c^2(2\nu + 1)$; the cofactor was found by polynomial division (SymPy `div`, exact, remainder 0) against the `ring_nf`-normalized goal, and is *replayed entirely inside Lean’s ring normalizer* — the computer-algebra system plays no trusted role. The full run history of this development — seven green build transcripts and the failed-attempt record, each committed the day it was produced with its diagnosis in the commit message; the certified companion script-and-transcript pair; and eight charter amendments recording every external audit round, every score checkpoint, and every method deviation (including the companion’s anchor deviation and depth correction, both registered *before* the certified run) — is committed under `scripts/` and `docs/C5-CHARTER.md`; per the repository’s audit protocol it is archival material and plays no role in the mathematics above.

Headline-theorem-to-artifact map (all in module `AmosClosure`; every row carries a direct oracle witness; the 74 inherited statements are mapped in [11, 12, 13]):

Result	Lean name	File	Status
family, positivity, Eq. (2)	amosFamily, amosFamily_calibration	AmosFamily.lean	exact
Eq. (3) + trichotomy	riccatiQReal_amosFamily etc.	AmosFamily.lean	exact
sufficiency half	amosFamily_upper_of_le_half	AmosFamily.lean	exact
ratio recurrence	besselIReal_ratio_recurrence	AmosLowerReal.lean	exact
lower bound L_ν	besselLowerReal_holds	AmosLowerReal.lean	exact
witness $x_0(c)$	amosFamily_lt_amosLower_witness	AmosLowerReal.lean	exact
Thm. 3.1	amosFamily_uniform_upper_iff	AmosLowerReal.lean	exact
Thm. 3.2	amosFamily_lower_of_one_le	AmosCrossing.lean	exact
Thm. 3.5	besselIReal_turan	AmosCrossing.lean	exact
window (§4.4)	amosFamily_crossing_exists	AmosCrossing.lean	exact
Eq. (4), threshold	crossing_hasDerivAt, crossing_threshold	AmosCrossing.lean	exact
uniqueness, orientation	crossing_unique, crossing_orientation_*	AmosCrossing.lean	exact
scale law	crossing_scale	AmosCrossing.lean	exact
Thm. 3.4	crossing_no_critical_contact	AmosTangency.lean	exact
Thm. 3.3 package	amosFamily_global_crossing	AmosTangency.lean	exact
§6 grid	c5_crossing_arb.py + transcript	scripts/	certified

From a clean clone at the release revision (on Windows, clone with `core.autocrlf=false` — LF bytes reproduce the companion script’s runtime self-hash — and `core.longpaths=true`):

```
lake exe cache get
lake build AmosClosure
lake env lean AmosClosure/Oracle.lean
python scripts/c5_crossing_arb.py
tectonic -X compile papers/c5-crossing/c5_crossing.tex
```

At the release revision, canonical artifact hashes will be recorded — tag-scoped, in `RELEASE-MANIFEST.md` in this paper’s directory — committed together with the release revision of this manuscript; compare against `git show <rev>:path`, never a worktree checkout. The annotated tag `c5-v1.0` will mark that release commit after the pre-release audits close, with the audit registry in `docs/C5-CHARTER.md`; the release-tree audit is recorded in the manifest, post-tag by sequencing, with any fixes in post-tag commits; tags never move.

9 Conclusion

For the diagonal Amos-type family at real order, the pinned formal development now contains the parameter classification — uniform upper bound iff $c \leq \frac{1}{2}$, uniform lower bound for $c \geq 1$, with explicit rational witnesses — and, in the intermediate regime $\frac{1}{2} < c < 1$, a global quantitative crossing theorem: exactly one crossing, transversal, strictly above an explicit threshold, with bilateral orientation, a constructive window, a two-sided scale law, and degenerate contact excluded by an exact curvature identity. Limitations: the in-core series definition only, and $\nu \geq 0$ (Section 5); the family, the classification as mathematics, and the crossing phenomenon are prior art [4] — the quantitative crossing package is the candidate-new content, stated with the priority prudence of Section 5. Natural next steps: the analogous classification for the second-kind ratio $K_{\nu+1}/K_\nu$ (where [6] suggests the tangency structure differs), asymptotics of $x_*(\nu, c)$ beyond the scale law, and upstreaming the modified-Bessel API.

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